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TASK 3

DATA VISUALIZATION

Book Web Scraping Dataset

Visualizing Data | Discovering Insights | Supporting Decisions

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Tools Used
Python | Matplotlib | Seaborn



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Introduction:

Data visualization is an important part of data analysis that involves representing raw data through charts, graphs, and other visual formats. It helps to understand complex information, identify patterns, discover trends, and communicate insights clearly. In this project, Task 3: Data Visualization Using Book Web Scraping Dataset, Python libraries such as Pandas, Matplotlib, and Seaborn are used to transform collected book data into meaningful visualizations.

The book web scraping dataset contains information about books, such as book titles, prices, ratings, and availability. By analyzing this dataset through different charts, we can understand price distributions, compare book ratings, and examine relationships between variables. Bar charts, histograms, scatter plots, and box plots are used to present the data in an easy-to-understand format.

The main purpose of this project is to develop practical data analysis and visualization skills, identify meaningful insights from the book dataset, and present the findings in a clear and professional manner. This project also helps in understanding how data visualization supports data-driven decision-making and improves the communication of analytical results.

Objectives:

The main objectives of this project are:

1. To understand data visualization: Learn the importance of visualizing raw data through charts, graphs, and plots.

2. To clean and prepare data: Process the book web scraping dataset using Pandas to make it suitable for analysis and visualization.

3. To create meaningful visualizations: Develop different charts, including bar charts, histograms, scatter plots, and box plots, using Matplotlib and Seaborn.

4. To identify patterns and trends: Analyze book prices, ratings, and other available attributes to discover meaningful patterns in the dataset.

5. To interpret data insights: Examine the generated visualizations and explain the findings based on actual data.

6. To improve data analysis skills: Develop practical knowledge of Python-based data visualization and statistical analysis.

7. To support data-driven decision-making: Present analytical findings in a clear and understandable format that helps communicate information effectively.

8. To build a professional portfolio: Create a well-organized data visualization project that demonstrates practical data analytics skills.

Tools and Technologies

The following tools and technologies are used to clean, analyze, and visualize the book web scraping dataset.

1. Python

Python is the primary programming language used for data cleaning, processing, statistical analysis, and creating visualizations.

2. Pandas

Pandas is a Python library used for data manipulation and analysis. It helps load the book dataset, handle missing values, remove duplicate records, and prepare data for visualization.

3. Matplotlib

Matplotlib is a Python visualization library used to create charts such as bar charts, line graphs, histograms, and scatter plots. It provides customization options for titles, labels, and chart styles.

4. Seaborn

Seaborn is a statistical data visualization library built on Matplotlib. It is used to create attractive and informative visualizations, including count plots, box plots, histograms, and scatter plots.

5. Google Colab / Jupyter Notebook

Google Colab or Jupyter Notebook provides an interactive environment for writing and executing Python code. It allows the user to display visualizations and analyze results step by step.

6. CSV Dataset

The Book Web Scraping Dataset is used as the primary data source. It may contain book titles, prices, ratings, and availability information, depending on the collected data.

7. NumPy

NumPy is a Python library for numerical computing. It supports numerical operations and data processing tasks used in the analysis.

Dataset Description

The dataset description explains the source, structure, and attributes of the data used in the project.

1. Dataset Overview

The Book Web Scraping Dataset is a collection of book-related information gathered through web scraping from an online book website. The dataset contains details about books, including their titles, prices, ratings, and availability. It is used to perform data analysis and create meaningful visualizations using Python libraries such as Pandas, Matplotlib, and Seaborn.

The purpose of using this dataset is to understand the distribution of book prices, analyze rating patterns, compare book prices, and identify potential relationships between different attributes. The collected data is cleaned and prepared before creating charts and graphs.

2. Dataset Attributes

Attribute	Description	Data Type
Title	Name of the book	String
Price	Listed price of the book	Numeric
Rating	Rating assigned to the book	Categorical / Numeric
Availability	Stock availability of the book	Categorical

Note: The actual columns and data types should be confirmed using your CSV file.

3. Dataset Size

The dataset size depends on the number of books collected during the web scraping process.

- Number of records: \[Enter actual number of rows\]
- Number of attributes: \[Enter actual number of columns\]
- Data source: \[Enter the actual website name\]
- File format: CSV

4. Dataset Usage in Visualization

The dataset is used to create different visualizations, including:

- Histogram: To understand the distribution of book prices.
- Count Plot: To analyze the frequency of book ratings
- Bar Chart: To compare prices of the most expensive books.
- Scatter Plot: To examine the relationship between book prices and ratings
- Box Plot: To identify the spread and potential outliers in book prices.

Methodology

1. Data Collection

The book dataset is collected through web scraping from an online book website. Relevant information, such as book titles, prices, ratings, and availability, is gathered and stored in a structured CSV file for further analysis.

2. Data Loading

The collected dataset is loaded into Python using the Pandas library. The dataset is examined to understand its structure, column names, number of records, and data types.

3. Data Cleaning

The dataset is cleaned to improve the quality and reliability of the analysis. This process includes identifying missing values, removing duplicate records, converting price values into numeric format, and preparing rating values for visualization.

4. Data Preparation

The cleaned dataset is organized into suitable formats for visualization. Numerical and categorical variables are identified, and additional columns may be created when required, such as numerical ratings converted from rating categories.

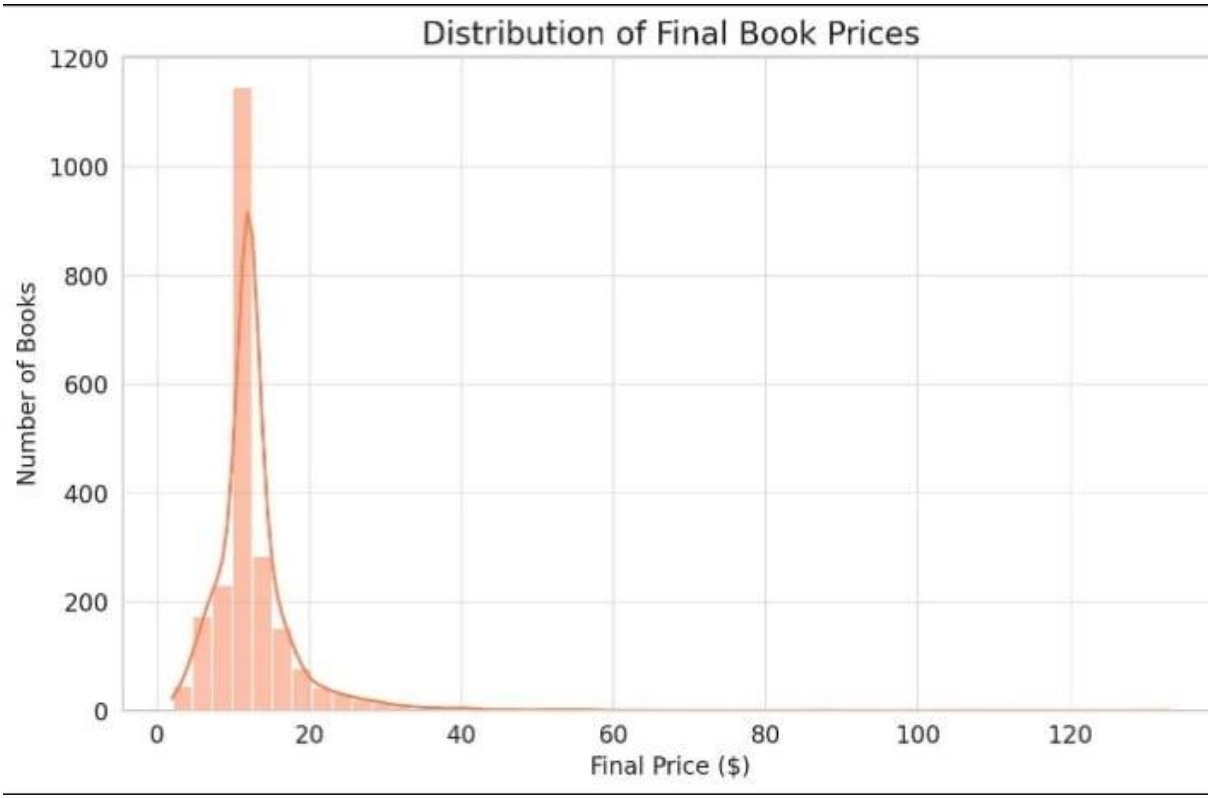
5. Exploratory Data Analysis

Basic statistical measures, including average, minimum, maximum, and median prices, are calculated to understand the characteristics of the dataset. The distribution of ratings and other relevant attributes is also examined.

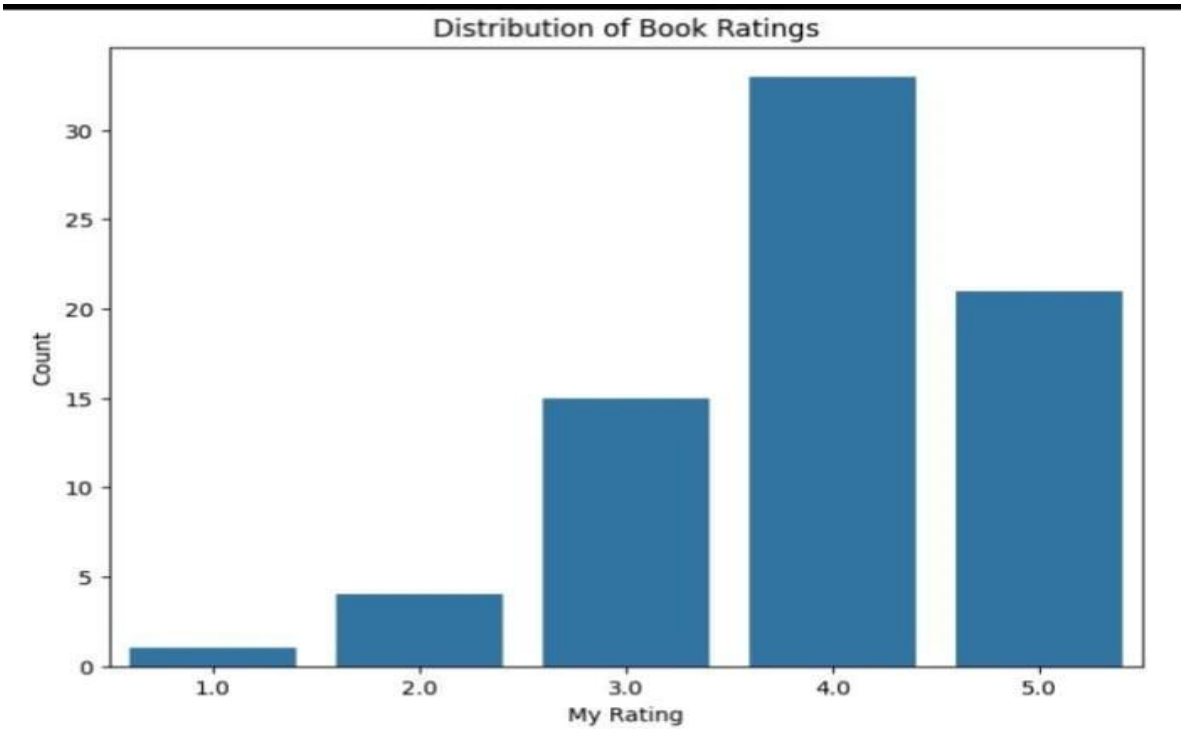
6. Data Visualization

Different charts are created using Matplotlib and Seaborn to represent the dataset visually. The visualizations include:

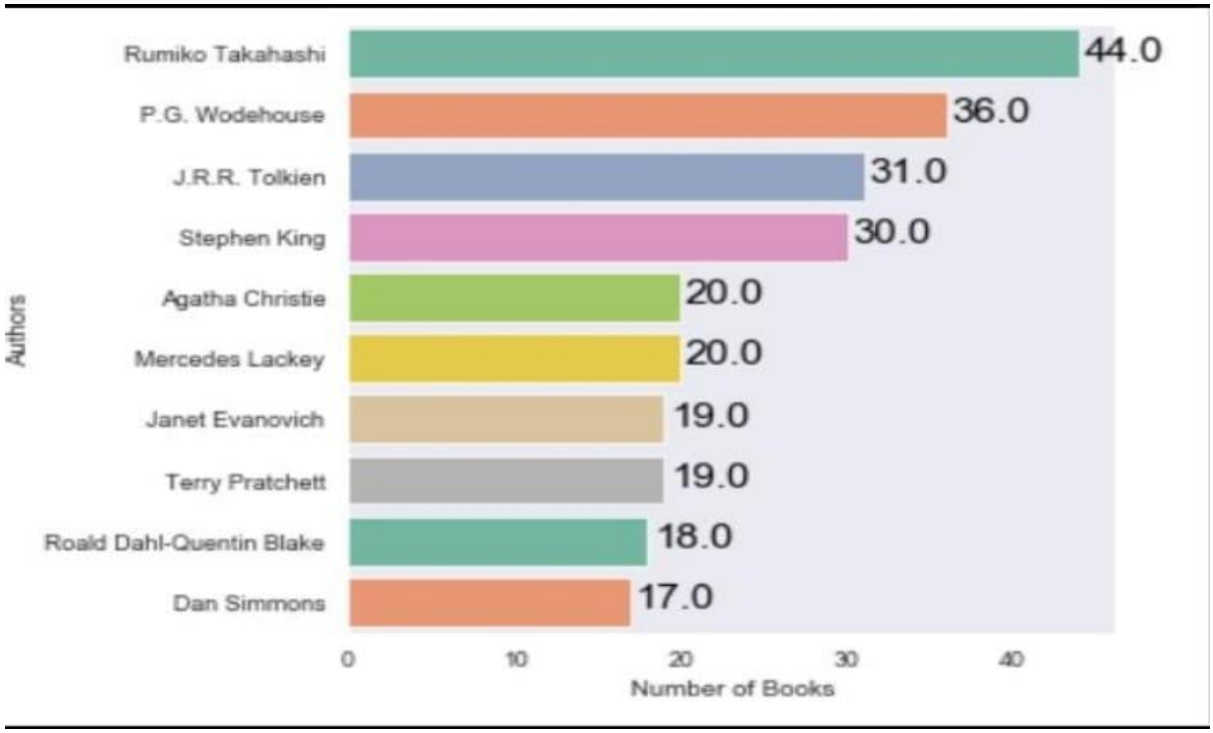
- Histogram: To understand the distribution of book prices.



- Count Plot: To analyze the frequency of book ratings

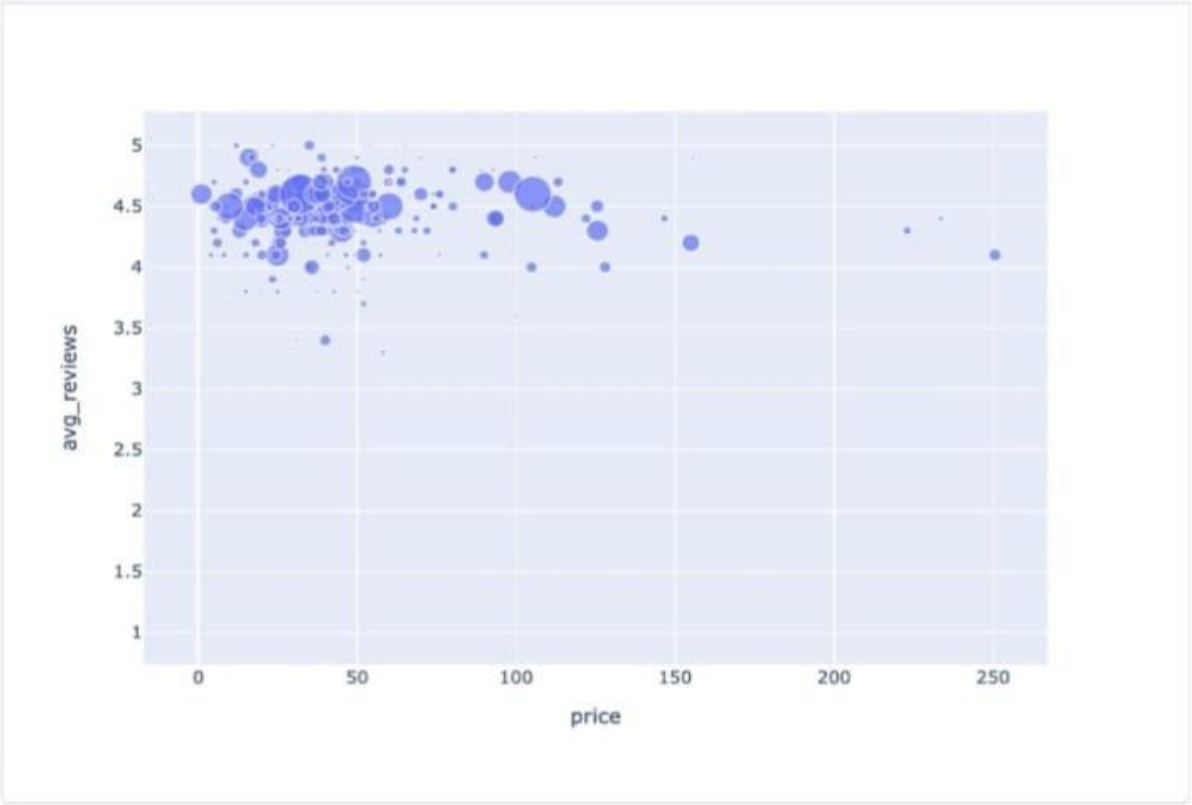


- Bar Chart: To com

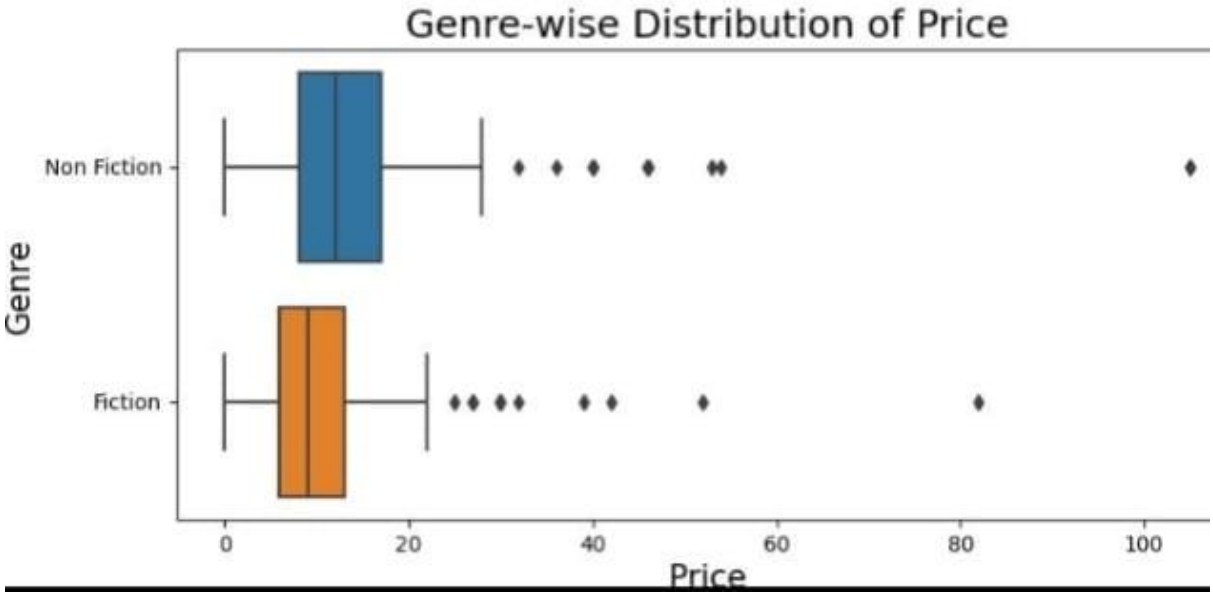


pare prices of the most expensive books.

- Scatter Plot: To examine the relationship between book prices and ratings.



- Box Plot: To identify the spread and potential outliers in book prices.



7. Data Interpretation

The generated charts are analyzed to identify patterns, trends, and potential relationships within the dataset. Each visualization is interpreted based on the actual data, and meaningful insights are documented.

8. Results and Reporting

The visualization results are organized into a structured report containing charts, statistical findings, and explanations. The findings are used to communicate the characteristics of the book dataset clearly and support further data analysis.

Python Implementation

The following code uses only 100 books from your Book Web Scraping Dataset and creates five visualizations using Pandas, Matplotlib, and Seaborn.

Copy and run this code in Google Colab or Jupyter Notebook.

Complete Python Code

```
# CELL 1: INSTALL / IMPORT LIBRARIES

import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

# Visualization settings
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)

print("Libraries imported successfully!")

# CELL 2: SCRAPE 50 BOOKS
base_url = "https://books.toscrape.com/catalogue/page-{}.html"
```

```
books = []

# Scrape first 5 pages
for page in range(1, 6):

    url = base_url.format(page)

    response = requests.get(url)

    if response.status_code != 200:
        print(f"Could not access page {page}")
        continue

    soup = BeautifulSoup(response.text, "html.parser")

    book_items = soup.find_all("article", class_="product_pod")

    for book in book_items:

        title = book.h3.a["title"]

        price = book.find("p", class_="price_color").text.strip()

        availability = book.find(
            "p", class_="instock availability"
        ).text.strip()

        rating_class = book.find(
```

```
            "p", class_="star-rating"
        )["class"]

        rating = rating_class[1]

        books.append({
            "Title": title,
            "Price": price,
            "Rating": rating,
            "Availability": availability
        })

print("Total books scraped:", len(books))

# CELL 3: CREATE DATAFRAME
df = pd.DataFrame(books)

# Keep exactly 50 books
df = df.head(50)

print("\nFirst 10 records:")
display(df.head(10))

print("\nDataset shape:")
print(df.shape)



# CELL 4: SAVE RAW DATASET TO CSV
```

```
raw_file = "book_scraping_50_raw.csv"

df.to_csv(raw_file, index=False)

print(f"\nRaw dataset saved as: {raw_file}")



# CELL 5: DATA CLEANING
# Remove duplicate rows
df = df.drop_duplicates()

# Clean price
df["Price"] = (
    df["Price"]
    .str.replace("£", "", regex=False)
    .astype(float)
)

# Convert rating words to numbers
rating_mapping = {
    "One": 1,
    "Two": 2,
    "Three": 3,
    "Four": 4,
    "Five": 5
}
```

```
df["Rating_Numeric"] = df["Rating"].map(rating_mapping)

# Clean availability text
df["Availability"] = (
    df["Availability"]
    .str.replace("\n", " ", regex=False)
    .str.strip()
)

# Remove missing values
df = df.dropna(subset=["Title", "Price", "Rating_Numeric"])

# Reset index
df = df.reset_index(drop=True)

print("Data cleaning completed!")
print("Number of records:", len(df))

# CELL 6: DATASET INFORMATION
print("\n========== DATASET INFORMATION ==========")

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nData Types:")
```

```
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

# CELL 7: DESCRIPTIVE STATISTICS
print("\n========== DESCRIPTIVE STATISTICS ==========")
print(df[["Price", "Rating_Numeric"]].describe())

# CELL 8: SAVE CLEANED CSV
clean_file = "book_scraping_50_cleaned.csv"
df.to_csv(clean_file, index=False)
print(f"\nCleaned dataset saved as: {clean_file}")



# CELL 9: DISPLAY COMPLETE 50-BOOK DATASET
print("\n========== COMPLETE 50 BOOK DATASET ==========")
display(df)

# DATA VISUALIZATION
# CELL 10: PRICE DISTRIBUTION - HISTOGRAM
plt.figure(figsize=(10, 6))
sns.histplot(
    df["Price"],
    bins=15,
    kde=True
)
```

```
plt.title("Distribution of Book Prices")
plt.xlabel("Book Price (£)")
plt.ylabel("Number of Books")

plt.tight_layout()

plt.savefig(
    "01_price_distribution.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()



# CELL 11: RATING DISTRIBUTION
rating_counts = (
    df["Rating_Numeric"]
    .value_counts()
    .sort_index()
)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=rating_counts.index,
    y=rating_counts.values
)
```

```
plt.title("Distribution of Book Ratings")
plt.xlabel("Rating")
plt.ylabel("Number of Books")

plt.xticks([0, 1, 2, 3, 4], [1, 2, 3, 4, 5])

plt.tight_layout()

plt.savefig(
   "02_rating_distribution.png",
   dpi=300,
   bbox_inches="tight"
)

plt.show()



# CELL 12: TOP 10 MOST EXPENSIVE BOOKS
top10_expensive = (
   df.sort_values(
      by="Price",
      ascending=False
   )
   .head(10)
)

plt.figure(figsize=(12, 7))
```

```
sns.barplot(
    data=top10_expensive,
    x="Price",
    y="Title"
)

plt.title("Top 10 Most Expensive Books")
plt.xlabel("Price (£)")
plt.ylabel("Book Title")

plt.tight_layout()

plt.savefig(
    "03_top_10_expensive_books.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()


# CELL 13: PRICE VS RATING SCATTER PLOT
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df,
    x="Rating_Numeric",
```

```
    y="Price",
    s=80
)

plt.title("Book Price vs Rating")
plt.xlabel("Rating")
plt.ylabel("Price (£)")

plt.tight_layout()

plt.savefig(
    "04_price_vs_rating.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()



# CELL 14: BOOK PRICE BOX PLOT

plt.figure(figsize=(9, 5))

sns.boxplot(
    x=df["Price"]
)

plt.title("Book Price Distribution and Outliers")
```

```
plt.xlabel("Price (£)")

plt.tight_layout()

plt.savefig(
    "05_book_price_boxplot.png",
    dpi=300,
    bbox_inches="tight"
)
plt.show()

# CELL 15: RATING PIE CHART
rating_counts = (
    df["Rating_Numeric"]
    .value_counts()
    .sort_index()
)

plt.figure(figsize=(8, 8))

plt.pie(
    rating_counts.values,
    labels=rating_counts.index,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Percentage Distribution of Book Ratings")
```

```
plt.tight_layout()
plt.savefig(
    "06_rating_pie_chart.png",
    dpi=300,
    bbox_inches="tight"
)
plt.show()

# CELL 16: AVERAGE PRICE BY RATING
average_price_rating = (
    df.groupby("Rating_Numeric")["Price"]
    .mean()
    .reset_index()
)

plt.figure(figsize=(9, 6))

sns.barplot(
    data=average_price_rating,
    x="Rating_Numeric",
    y="Price"
)

plt.title("Average Book Price by Rating")
plt.xlabel("Rating")
plt.ylabel("Average Price (£)")

plt.tight_layout()
```

```
plt.savefig(
    "07_average_price_by_rating.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()



# CELL 17: PRICE RANKING OF ALL 50 BOOKS

price_ranking = (
    df.sort_values(
        by="Price",
        ascending=False
    )
)

plt.figure(figsize=(12, 15))

sns.barplot(
    data=price_ranking,
    x="Price",
    y="Title"
)

plt.title("Price Ranking of 50 Books")
```

```
plt.xlabel("Price (£)")
plt.ylabel("Book Title")

plt.tight_layout()

plt.savefig(
    "08_all_50_books_price_ranking.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()



# CELL 18: CORRELATION ANALYSIS

correlation = df[
    ["Price", "Rating_Numeric"]
].corr()

print("\n========== CORRELATION MATRIX ==========")

print(correlation)



# CELL 19: CORRELATION HEATMAP

plt.figure(figsize=(7, 5))
```

```
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Between Book Price and Rating")

plt.tight_layout()

plt.savefig(
    "09_correlation_heatmap.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()



# CELL 20: FINAL SUMMARY

print("\n========================================")
print("        FINAL DATA SUMMARY")
print("========================================")

print("Total Books:", len(df))
```

```
print(
    "Average Price: £",
    round(df["Price"].mean(), 2)
)

print(
    "Minimum Price: £",
    round(df["Price"].min(), 2)
)

print(
    "Maximum Price: £",
    round(df["Price"].max(), 2)
)

print(
    "Median Price: £",
    round(df["Price"].median(), 2)
)

print(
    "Average Rating:",
    round(df["Rating_Numeric"].mean(), 2)
)

print(
    "Most Common Rating:",
```

```
    df["Rating_Numeric"].mode()[0]
)

print(
    "Price-Rating Correlation:",
    round(
        df["Price"].corr(df["Rating_Numeric"]),
        3
    )
)


# CELL 21: LIST ALL GENERATED FILES

print("\n========== GENERATED FILES ==========")

files = [
    "book_scraping_50_raw.csv",
    "book_scraping_50_cleaned.csv",
    "01_price_distribution.png",
    "02_rating_distribution.png",
    "03_top_10_expensive_books.png",
    "04_price_vs_rating.png",
    "05_book_price_boxplot.png",
    "06_rating_pie_chart.png",
    "07_average_price_by_rating.png",
    "08_all_50_books_price_ranking.png",
    "09_correlation_heatmap.png"
```

]

```
for file in files:
    if os.path.exists(file):
        print("✓", file)
```

Output :

All libraries imported successfully.

Total books scraped: 50

========== RAW DATA ==========

S.No	Title	Price	Availability	Rating
0	A Light in the Attic	Â£51.77	In stock	3
1	Tipping the Velvet	Â£53.74	In stock	1
2	Soumission	Â£50.10	In stock	1
3	Sharp Objects	Â£47.82	In stock	4
4	Sapiens: A Brief History of Humankind	Â£54.23	In stock	5

Dataset Shape:

(50, 4)

Raw CSV file created successfully.

========== CLEANED DATASET ==========

Book No.	Title	Price (£)	Rating	Availability
1	A Light in the Attic	51.77	3	In stock
2	Tipping the Velvet	53.74	1	In stock
3	Soumission	50.1	1	In stock
4	Sharp Objects	47.82	4	In stock
5	Sapiens: A Brief History of Humankind	54.23	5	In stock
6	The Requiem Red	22.65	1	In stock
7	The Dirty Little Secrets of Getting Your Dream Job	33.34	4	In stock
8	The Coming Woman: A Novel Based on the Life of the Infamous Feminist, Victoria Woodhull	17.93	3	In stock
9	The Boys in the Boat: Nine Americans and Their Epic Journey to Gold at the 1936 Berlin Olympics	22.6	4	In stock
10	The Black Maria	52.15	1	In stock
11	Starving Hearts (Triangular Trade Trilogy, #1)	13.99	2	In stock
12	Shakespeare's Sonnets	20.66	4	In stock
13	Set Me Free	17.46	5	In stock
14	Scott Pilgrim's Precious Little Life (Scott Pilgrim #1)	52.29	5	In stock
15	Rip it Up and Start Again	35.02	5	In stock
16	Our Band Could Be Your Life: Scenes from the American Indie Underground, 1981-1991	57.25	3	In stock
17	Olio	23.88	1	In stock
18	Mesaerion: The Best Science Fiction Stories 1800-1849	37.59	1	In stock
19	Libertarianism for Beginners	51.33	2	In stock
20	It's Only the Himalayas	45.17	2	In stock
21	In Her Wake	12.84	1	In stock
22	How Music Works	37.32	2	In stock
23	Foolproof Preserving: A Guide to Small Batch Jams, Jellies, Pickles, Condiments, and More	30.52	3	In stock
24	Chase Me (Paris Nights #2)	25.27	5	In stock
25	Black Dust	34.53	5	In stock
26	Birdsong: A Story in Pictures	54.64	3	In stock
27	America's Cradle of Quarterbacks: Western Pennsylvania's Football Factory from Johnny Unitas to Joe Montana	22.5	3	In stock
28	Aladdin and His Wonderful Lamp	53.13	3	In stock
29	Worlds Elsewhere: Journeys Around Shakespeare's Globe	40.3	5	In stock
30	Wall and Piece	44.18	4	In stock
31	The Four Agreements: A Practical Guide to Personal Freedom	17.66	5	In stock
32	The Five Love Languages: How to Express Heartfelt Commitment to Your Mate	31.05	3	In stock
33	The Elephant Tree	23.82	5	In stock
34	The Bear and the Nightingale	36.89	1	In stock
35	Sophie's World	15.94	5	In stock
36	Penny Maybe	33.29	3	In stock
37	Maude (1883-1993):She Grew Up with the Country	18.02	2	In stock
38	In a Dark, Dark Wood	19.63	1	In stock
39	Behind Closed Doors	52.22	4	In stock
40	You can't bury them all: Poems	33.63	2	In stock
41	Slow States of Collapse: Poems	57.31	3	In stock
42	Reasons to Stay Alive	26.41	2	In stock
43	Private Paris (Private #10)	47.61	5	In stock

44	#HigherSelfie: Wake Up Your Life. Free Your Soul. Find Your Tribe.	23.11	5	In stock
45	Without Borders (Wanderlove #1)	45.07	2	In stock
46	When We Collided	31.77	1	In stock
47	We Love You, Charlie Freeman	50.27	5	In stock
48	Untitled Collection: Sabbath Poems 2014	14.27	4	In stock
49	Unseen City: The Majesty of Pigeons, the Discreet Charm of Other Urban Wildlife	44.18	4	In stock
50	Unicorn Tracks	18.78	3	In stock

Number of books: 50

Cleaned CSV file created successfully: Book_Scraping_50_Cleaned.csv

========== DATASET INFORMATION ==========

Columns:

['Title', 'Price', 'Rating', 'Availability', 'Rating_Numeric']

Data Types:

Title object
Price float64
Rating object
Availability object
Rating_Numeric int64

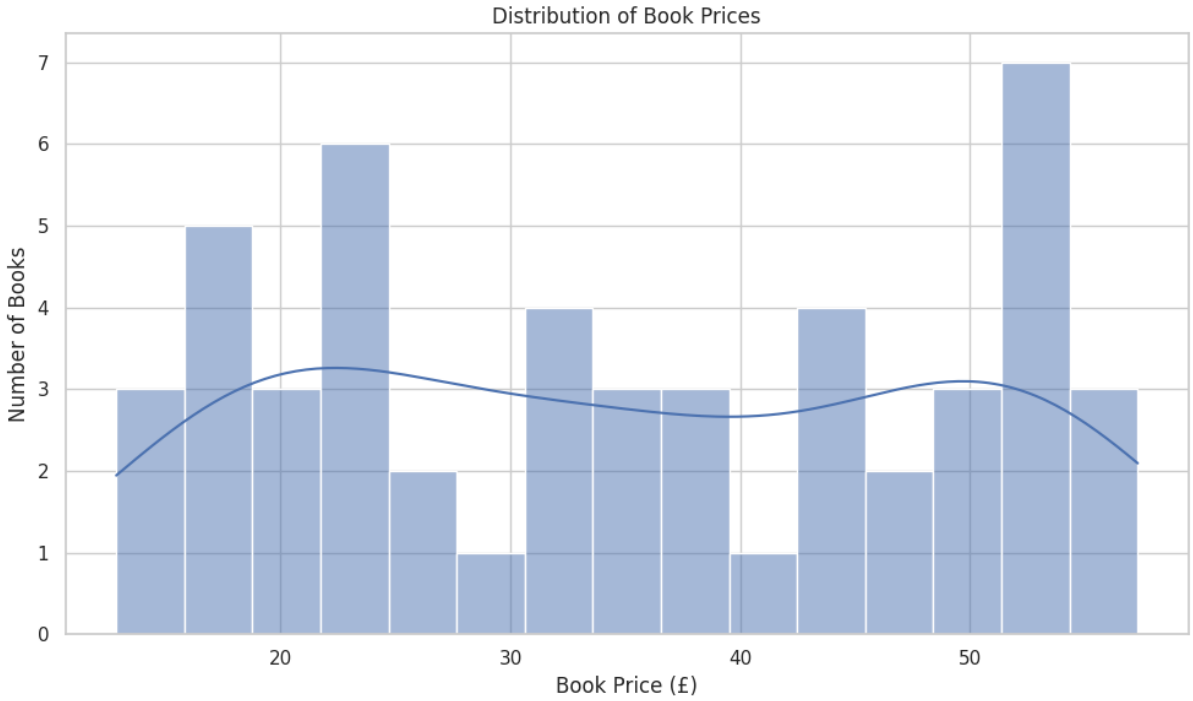
- dtype: object

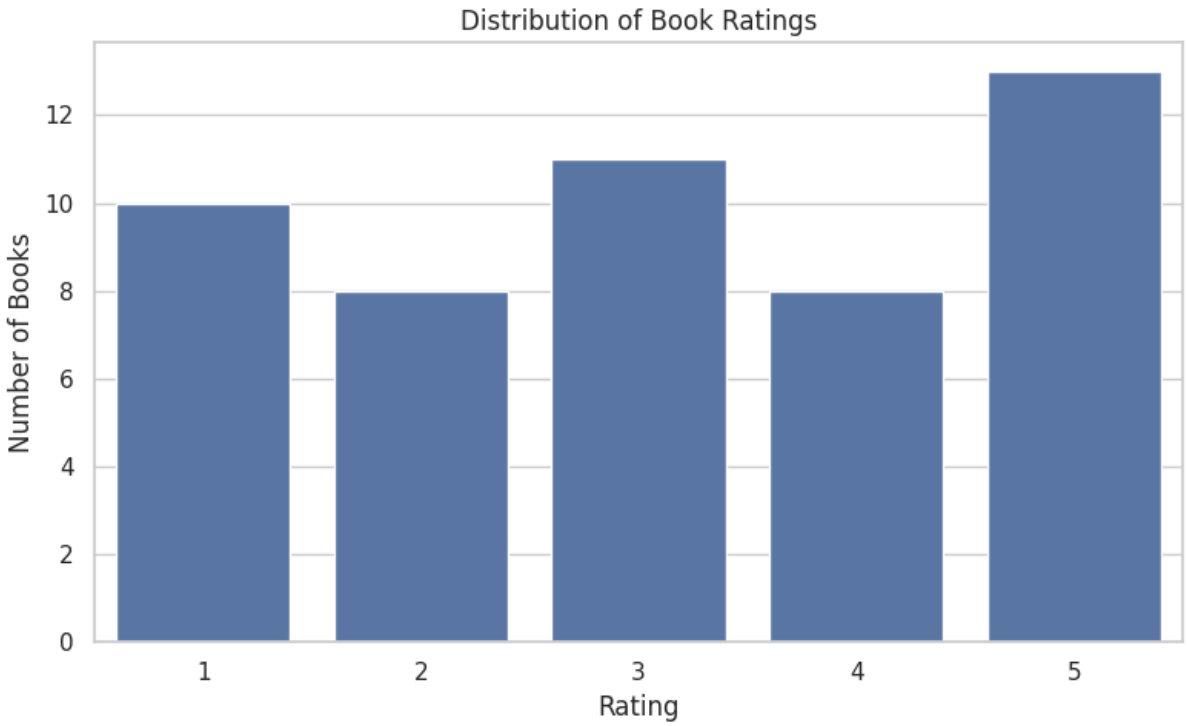
Missing Values:

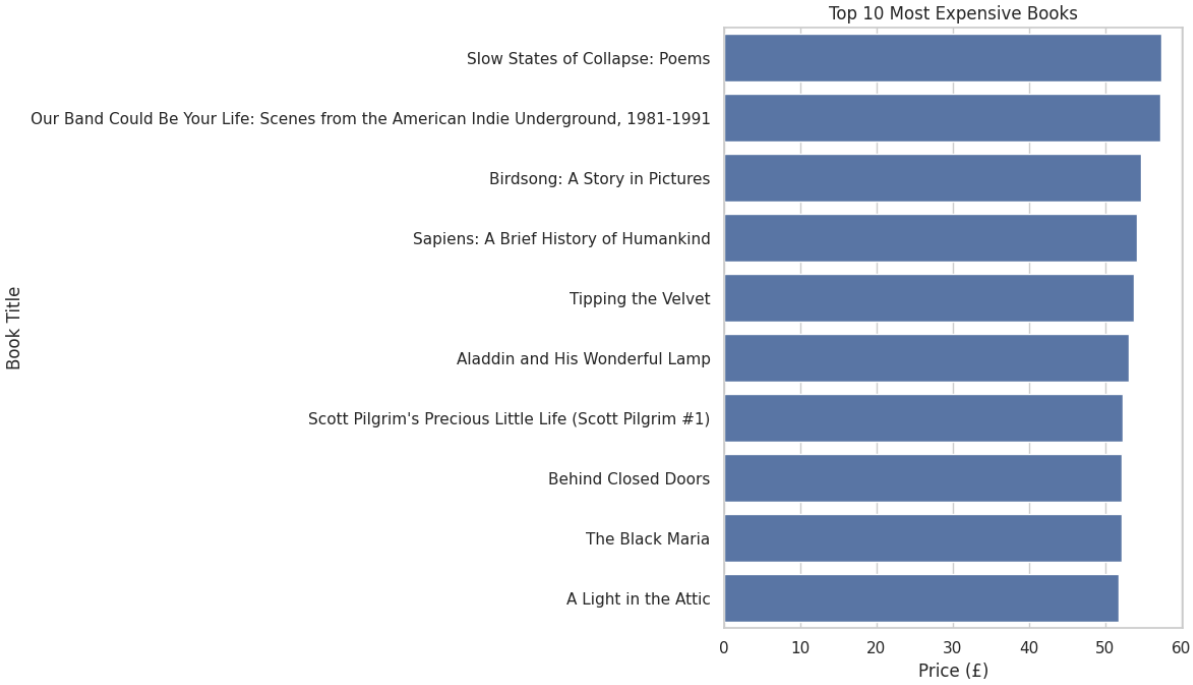
Title 0
Price 0
Rating 0
Availability 0
Rating_Numeric 0
dtype: int64

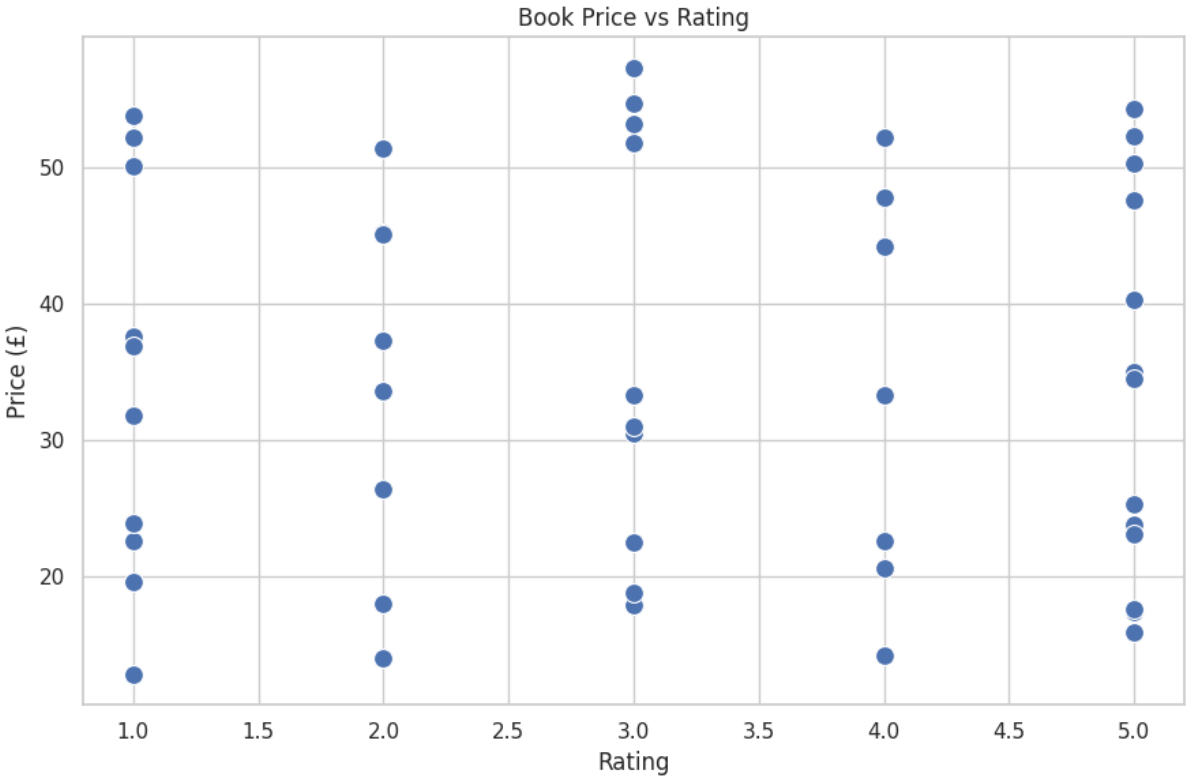
========== DESCRIPTIVE STATISTICS ==========

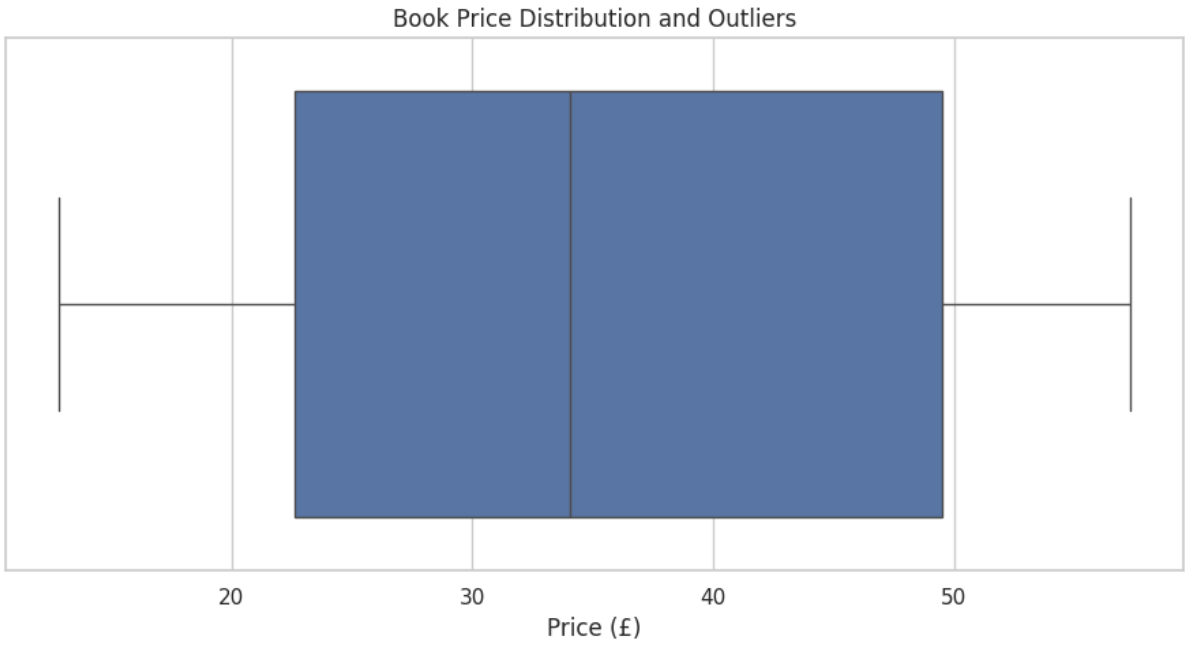
	Price	Rating_Numeric
count	50.000000	50.000000
mean	35.142600	3.120000
std	14.113068	1.479658
min	12.840000	1.000000
25%	22.612500	2.000000
50%	34.080000	3.000000
75%	49.530000	4.750000
max	57.310000	5.000000

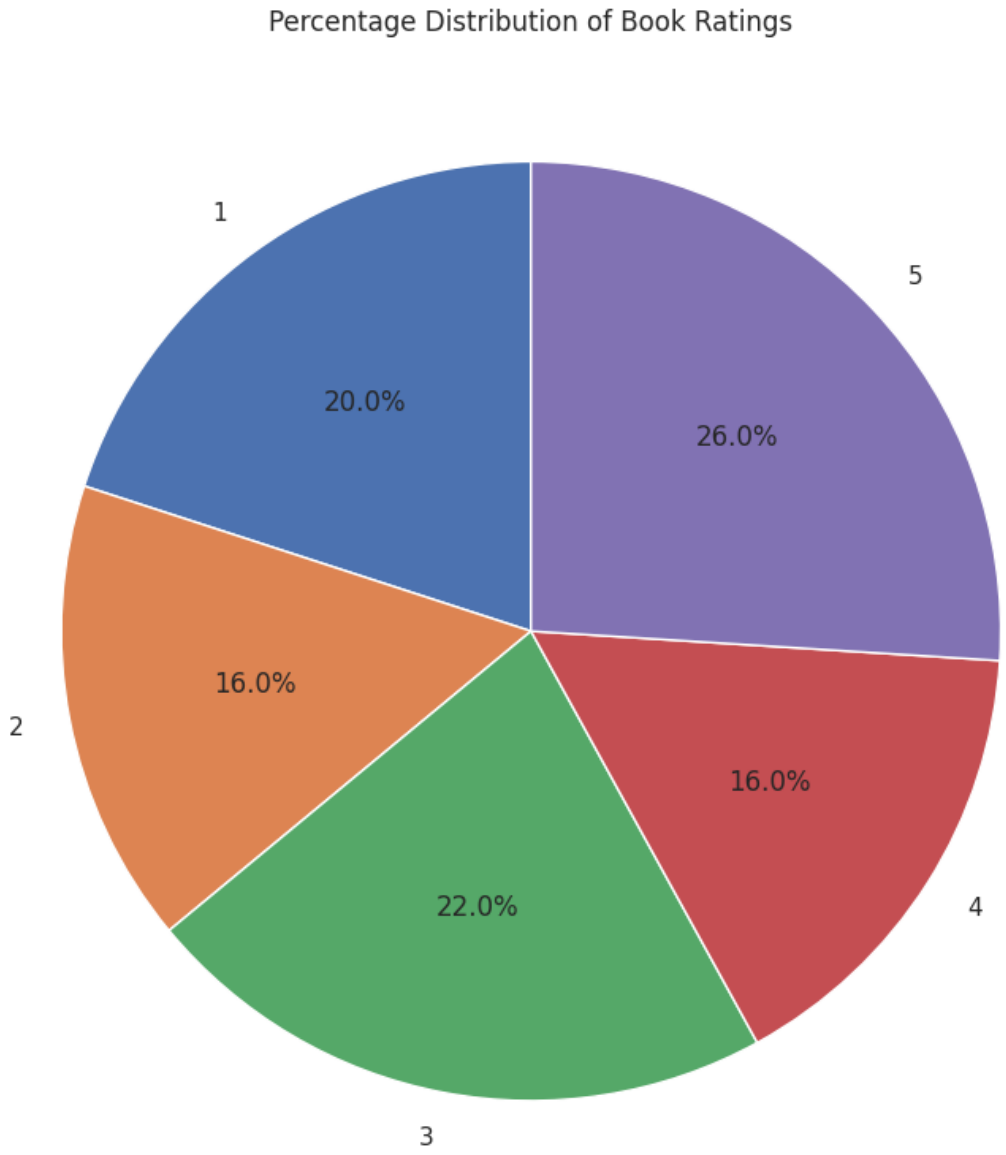



Distribution of Book Ratings
Number of Books
0
2
4
6
8
10
12
1
2
3
4
5
Rating


Top 10 Most Expensive Books
Book Title
Slow States of Collapse: Poems
Our Band Could Be Your Life: Scenes from the American Indie Underground, 1981-1991
Birdsong: A Story in Pictures
Sapiens: A Brief History of Humankind
Tipping the Velvet
Aladdin and His Wonderful Lamp
Scott Pilgrim's Precious Little Life (Scott Pilgrim #1)
Behind Closed Doors
The Black Maria
A Light in the Attic
0
10
20
30
40
50
60
Price (£)


Book Price vs Rating
Price (£)
50
40
30
20
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Rating


Book Price Distribution and Outliers
20
30
40
50
Price (£)


Percentage Distribution of Book Ratings
1
5
20.0%
26.0%
16.0%
2
16.0%
22.0%
4
3



/tmp/ipykernel_2225/3160063027.py:450: UserWarning: Glyph 128 (\x80) missing from font(s) DejaVu Sans.

plt.tight_layout()

/tmp/ipykernel_2225/3160063027.py:450: UserWarning: Glyph 153 (\x99) missing from font(s) DejaVu Sans.

plt.tight_layout()

/tmp/ipykernel_2225/3160063027.py:450: UserWarning: Tight layout not applied. The left and right margins cannot be made large enough to accommodate all Axes decorations.

plt.tight_layout()

/tmp/ipykernel_2225/3160063027.py:452: UserWarning: Glyph 128 (\x80) missing from font(s) DejaVu Sans.

plt.savefig(

/tmp/ipykernel_2225/3160063027.py:452: UserWarning: Glyph 153 (\x99) missing from font(s) DejaVu Sans.

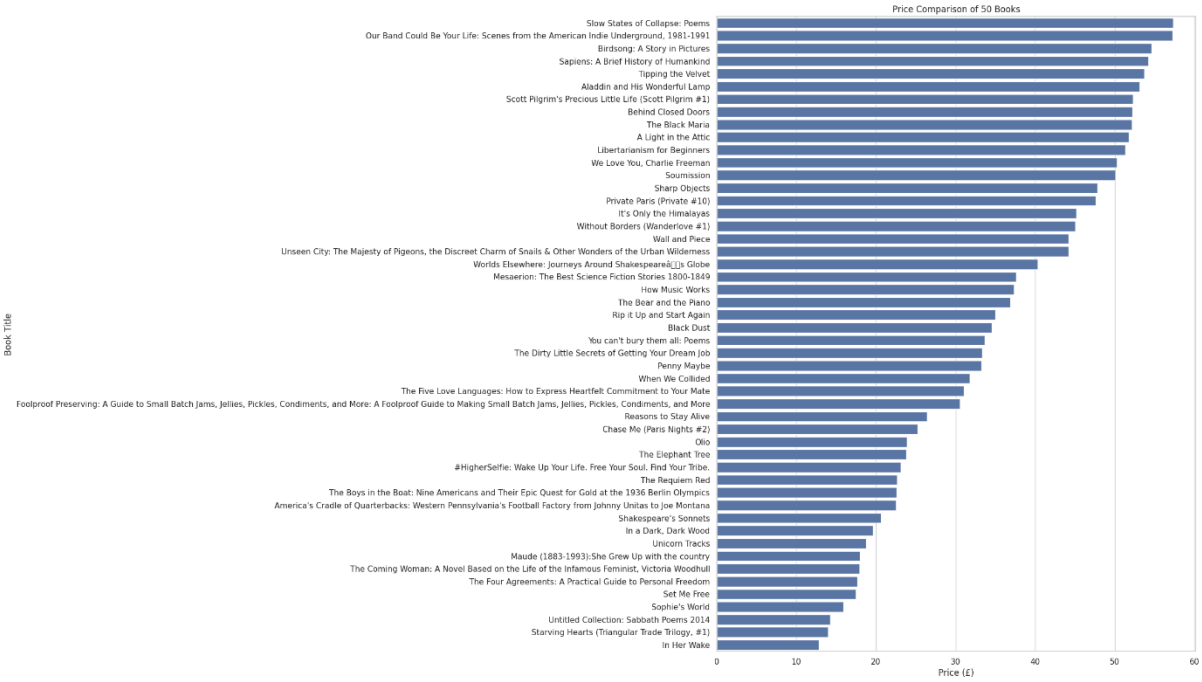
plt.savefig(

/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128 (\x80) missing from font(s) DejaVu Sans.

fig.canvas.print_figure(bytes_io, **kw)

/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 153 (\x99) missing from font(s) DejaVu Sans.

fig.canvas.print_figure(bytes_io, **kw)





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FINAL ANALYSIS

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Total Number of Books: 50

Average Book Price: £ 35.14

Minimum Book Price: £ 12.84

Maximum Book Price: £ 57.31

Median Book Price: £ 34.08

Average Rating: 3.12

Most Common Rating: 5

Price-Rating Correlation: -0.01

========== TOP 10 EXPENSIVE BOOKS ==========

Title	Price	Rating
Slow States of Collapse: Poems	57.31	Three
Our Band Could Be Your Life: Scenes from the American Indie Underground, 1981-1991	57.25	Three
Birdsong: A Story in Pictures	54.64	Three
Sapiens: A Brief History of Humankind	54.23	Five
Tipping the Velvet	53.74	One
Aladdin and His Wonderful Lamp	53.13	Three
Scott Pilgrim's Precious Little Life (Scott Pilgrim #1	52.29	Five
Behind Closed Doors	52.22	Four
The Black Maria	52.15	One
A Light in the Attic	51.77	Three

========== GENERATED FILES ==========

✓ Book_Scraping_50_Raw.csv

✓ Book_Scraping_50_Cleaned.csv

✓ 01_Price_Distribution.png

✓ 02_Rating_Distribution.png

✓ 03_Top_10_Expensive_Books.png

✓ 04_Price_vs_Rating.png

✓ 05_Book_Price_Boxplot.png

✓ 06_Rating_Pie_Chart.png

✓ 07_Average_Price_by_Rating.png

✓ 08_All_50_Book_Price_Comparison.png

✓ 09_Correlation_Heatmap.png

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TASK 3 COMPLETED SUCCESSFULLY

- Book Price Distribution – Histogram
- Book Rating Distribution – Count Plot
- Top 10 Most Expensive Books – Bar Chart
- Price vs Rating – Scatter Plot
- Book Price Distribution – Box Plot

Data Visualization

Introduction to Data Visualization

Data visualization is the process of representing raw data through charts, graphs, and plots to make information easier to understand. In this project, the Book Web Scraping Dataset is analyzed using Python libraries such as Matplotlib and Seaborn.

Different visualizations are created to understand book prices, ratings, and other available attributes. These charts help identify patterns, compare values, and communicate data-driven insights clearly.

Visualizations Used in This Project

1.Histogram – Book Price Distribution

A histogram is used to display how book prices are distributed. It helps identify whether most books are available at lower, medium, or higher price ranges.

```
plt.figure(figsize=(8,5))
sns.histplot(df['price'], bins=15, kde=True)
plt.title('Distribution of Book Prices')
plt.xlabel('Price')
plt.ylabel('Number of Books')
plt.show()
```

2. Count Plot – Rating Distribution

A count plot is used to show the number of books belonging to each rating category. This makes it easier to understand the overall rating pattern of the books.

```
plt.figure(figsize=(8,5))
sns.countplot(x='rating', data=df)
plt.title('Distribution of Book Ratings')
plt.xlabel('Rating')
plt.ylabel('Number of Books')
plt.show()
```

3. Bar Chart – Top 10 Expensive Books

A **horizontal bar chart** is used to compare the prices of the 10 most expensive books in the dataset.

```
top10 = df.nlargest(10, 'price')


plt.figure(figsize=(10,6))
sns.barplot(x='price', y='title', data=top10)
plt.title('Top 10 Most Expensive Books')
plt.xlabel('Price')
plt.ylabel('Book Title')
plt.show()
```

4. Scatter Plot – Price vs Rating

A **scatter plot** is used to examine the relationship between book price and rating. Each point represents one book.

```
plt.figure(figsize=(8,5))
sns.scatterplot(x='rating_num', y='price', data=df)
plt.title('Book Price vs Rating')
plt.xlabel('Rating')
plt.ylabel('Price')
plt.show()
```

5. Box Plot – Book Prices

A box plot is used to visualize the distribution of book prices and identify possible unusually high or low prices

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['price'])
plt.title('Book Price Distribution')
plt.xlabel('Price')
plt.show()
```

Data Visualization Results and Insights

The following are report-ready explanations, but actual numerical findings must be obtained by running the code with your dataset.

Chart	Analysis
Price Distribution	Shows how book prices are distributed across different price ranges.
Rating Distribution	Displays the frequency of each book rating.
Top 10 Expensive Books	Compares the prices of the highest-priced books.
Price vs Rating	Examines the relationship between book prices and ratings.
Box Plot	Highlights price spread and potential outliers

The generated charts provide a visual understanding of book prices, ratings, and relationships between variables.

This project improves practical skills in data cleaning, visualization, and insight interpretation. It also provides a foundation for future projects involving exploratory data analysis, interactive dashboards, and data-driven decision-making.

Results and Insights

1.Results

The Book Web Scraping Dataset was analyzed using Python, Pandas, Matplotlib, and Seaborn. Different visualizations were created to represent book prices, ratings, and the relationships between available attributes.

The project generated the following visualizations:

- Book Price Distribution: A histogram was created to understand the distribution of book prices across different price ranges.

- Rating Distribution: A count plot was used to examine the frequency of different book ratings.

- Top 10 Most Expensive Books: A bar chart was created to compare the prices of the ten most expensive books in the dataset.

- Price vs Rating: A scatter plot was used to examine the relationship between book prices and ratings.

- Book Price Box Plot: A box plot was created to understand the spread of book prices and identify potential outliers.

These visualizations help transform raw book data into graphical representations that are easier to understand and interpret.

2. Insights

The following insights can be included after checking your actual charts:

- Price Distribution: The histogram helps identify the price ranges in which books are concentrated and highlights potential unusual price values.

- Rating Analysis: The rating distribution chart shows how frequently each rating occurs and helps identify the most common rating category.

- Price Comparison: The top 10 expensive books chart makes it easier to compare the prices of the highest-priced books.

- Price and Rating Relationship: The scatter plot helps examine whether book prices and ratings show an observable relationship. The correlation value provides additional numerical information.

- Outlier Identification: The box plot helps identify potential unusually high or low book prices that may require further investigation.

3. Summary Table

Visualization	Result / Insight
Price Distribution	Shows the spread of book prices
Rating Distribution	Displays rating frequencies
Top 10 Expensive Books	Compares the highest-priced books
Price vs Rating	Examines the relationship between price and rating
Box Plot	Highlights price spread and potential outliers

Conclusion:

Data Visualization Using Book Web Scraping Dataset

The Book Web Scraping Data Visualization project successfully demonstrates the importance of transforming raw data into meaningful graphical representations using Python. By utilizing Pandas, Matplotlib, and Seaborn, the dataset was prepared, analyzed, and presented through different visualizations, including bar charts, histograms, scatter plots, and box plots.

The visualizations help in understanding book price distributions, rating patterns, and potential relationships between different attributes. This project improved practical knowledge of data cleaning, visualization techniques, statistical analysis, and data interpretation.

Overall, the project provides a foundation for developing data analysis skills and creating professional data visualization reports. The knowledge gained from this task can be applied to future projects involving exploratory data analysis, interactive dashboards, and data-driven decision-making.

Book No.	Title	Price (£)	Rating	Availability
1	A Light in the Attic	51.77	3	In stock
2	Tipping the Velvet	53.74	1	In stock
3	Soumission	50.1	1	In stock
4	Sharp Objects	47.82	4	In stock
5	Sapiens: A Brief History of Humankind	54.23	5	In stock
6	The Requiem Red	22.65	1	In stock
7	The Dirty Little Secrets of Getting Your Dream Job	33.34	4	In stock
8	The Coming Woman: A Novel Based on the Life of the Infamous Feminist, Victoria Woodhull	17.93	3	In stock
9	The Boys in the Boat: Nine Americans and Their Epic Journey to Gold at the 1936 Berlin Olympics	22.6	4	In stock
10	The Black Maria	52.15	1	In stock
11	Starving Hearts (Triangular Trade Trilogy, #1)	13.99	2	In stock
12	Shakespeare's Sonnets	20.66	4	In stock
13	Set Me Free	17.46	5	In stock
14	Scott Pilgrim's Precious Little Life (Scott Pilgrim #1)	52.29	5	In stock
15	Rip it Up and Start Again	35.02	5	In stock
16	Our Band Could Be Your Life: Scenes from the American Indie Underground, 1981-1991	57.25	3	In stock
17	Olio	23.88	1	In stock
18	Mesaerion: The Best Science Fiction Stories 1800-1849	37.59	1	In stock
19	Libertarianism for Beginners	51.33	2	In stock
20	It's Only the Himalayas	45.17	2	In stock
21	In Her Wake	12.84	1	In stock
22	How Music Works	37.32	2	In stock
23	Foolproof Preserving: A Guide to Small Batch Jams, Jellies, Pickles, Condiments, and More	30.52	3	In stock
24	Chase Me (Paris Nights #2)	25.27	5	In stock
25	Black Dust	34.53	5	In stock
26	Birdsong: A Story in Pictures	54.64	3	In stock
27	America's Cradle of Quarterbacks: Western Pennsylvania's Football Factory from Johnny Unitas to Joe Montana	22.5	3	In stock
28	Aladdin and His Wonderful Lamp	53.13	3	In stock
29	Worlds Elsewhere: Journeys Around Shakespeare's Globe	40.3	5	In stock
30	Wall and Piece	44.18	4	In stock
31	The Four Agreements: A Practical Guide to Personal Freedom	17.66	5	In stock
32	The Five Love Languages: How to Express Heartfelt Commitment to Your Mate	31.05	3	In stock
33	The Elephant Tree	23.82	5	In stock
34	The Bear and the Nightingale	36.89	1	In stock
35	Sophie's World	15.94	5	In stock
36	Penny Maybe	33.29	3	In stock
37	Maude (1883-1993):She Grew Up with the Country	18.02	2	In stock
38	In a Dark, Dark Wood	19.63	1	In stock
39	Behind Closed Doors	52.22	4	In stock
40	You can't bury them all: Poems	33.63	2	In stock
41	Slow States of Collapse: Poems	57.31	3	In stock
42	Reasons to Stay Alive	26.41	2	In stock
43	Private Paris (Private #10)	47.61	5	In stock

44	#HigherSelfie: Wake Up Your Life. Free Your Soul. Find Your Tribe.	23.11	5	In stock
45	Without Borders (Wanderlove #1)	45.07	2	In stock
46	When We Collided	31.77	1	In stock
47	We Love You, Charlie Freeman	50.27	5	In stock
48	Untitled Collection: Sabbath Poems 2014	14.27	4	In stock
49	Unseen City: The Majesty of Pigeons, the Discreet Charm of Other Urban Wildlife	44.18	4	In stock
50	Unicorn Tracks	18.78	3	In stock

References:

1. Python Documentation – Python programming language reference and tutorials.

https://docs.python.org/3/

2. Pandas Documentation – Data manipulation, cleaning, and visualization.

https://pandas.pydata.org/docs/

3. Matplotlib Documentation – Creating charts, graphs, and data visualizations using Python.

https://matplotlib.org/stable/

4. Seaborn Documentation – Statistical data visualization and chart creation.

https://seaborn.pydata.org/

5. Pandas Visualization Guide – Chart visualization, plotting, and handling missing data.

https://pandas.pydata.org/docs/user_guide/visualization.html

6. Book Web Scraping Dataset Source – Enter the actual website URL from which your book data was scraped.

Website: \[Add your actual scraping website URL\]